CHAPTER

# 12

# Single Nucleotide Polymorphisms and Applications

Core short tandem repeat (STR) loci are being used extensively today and will probably continue to be used for many years in the future because they are part of the DNA databases that are growing around the world. Yet forensic DNA scientists often use additional markers as the need arises to obtain further information about a particular sample. This chapter will focus on efforts with single nucleotide polymorphisms (SNPs) and insertion-deletion (indel) biallelic polymorphisms and their applications including ancestry estimation and phenotype prediction.

## SINGLE NUCLEOTIDE POLYMORPHISMS (SNPs)

A single-base sequence variation between individuals at a particular point in the genome is often referred to as a *single nucleotide polymorphism*, or SNP (pronounced "snip"). SNPs are abundant in the human genome and, as such, are being used for linkage studies to track genetic diseases (Brookes 1999). Millions of SNPs exist per individual and thus the abundance of SNPs means that they could be used to help differentiate individuals from one another. Table 12.1 compares and contrasts SNP and STR markers.

A number of technologies have been developed to minaturize and automate the procedure for SNP analysis. Millions of SNPs can be processed simultaneously using next-generation sequencing technologies (Ku et al. 2010) from companies such as Illumina (San Diego, CA).

### Pros and Cons of SNPs

SNPs have been considered as potential genetic markers by the forensic DNA community for several reasons. First and foremost, the polymerase chain reaction (PCR) products from

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**TABLE 12.1** Comparison of STR and SNP Markers. SNPs Are More Common in the Human Genome Than STRs But Are Not as Polymorphic.

| Characteristics | Short Tandem Repeats (STRs) | Single Nucleotide Polymorphisms (SNPs) |
|---|---|---|
| Occurrence in human genome | ≈1 in every 15 kb | ≈1 in every 1 kb |
| General informativeness | High | Low; only 20% to 30% as informative as STRs |
| Mutation rate | ≈1 in 1000 | ≈1 in 100 000 000 |
| Marker type | Di-, tri-, tetra-, penta-nucleotide repeat markers with many alleles | Mostly bi-allelic markers with six possibilities: A/G, C/T, A/T, C/G, T/G, A/C |
| Number of alleles per marker | Usually 5 to 20 | Typically 2 (some tri-allelic SNPs exist) |
| Detection methods | Gel/capillary electrophoresis | Sequence analysis; microchip hybridization |
| Multiplex capability | > 10 markers with multiple fluorescent dyes | Difficult to amplify more than 50 SNPs well (detection of 1000s with microchips) |
| Amplicon size | ≈75 to 400 bp | Can be less than 100 bp |
| Ability to predict ethnicity (biogeographical ancestry) | Limited | Some SNPs associated with ethnicity |
| Phenotypic information | No | Possible to predict some hair colors, etc. |
| Major advantages for forensic application | Many alleles enabling higher success rates for detecting and deciphering mixtures | PCR products can be made small potentially enabling higher success rates with degraded DNA samples; low mutation rate may aid kinship analysis; phenotype prediction |
| Limitations for forensic application | Data interpretation must account for artifacts such as dye blobs, stutter, spikes, etc. | No widely established core loci; large multiplexing assays required; mixture resolution issues/interpretation; multiple typing platforms make universal SNP selection difficult; population substructure due to low mutation rate |

SNPs can be less than 100 bp in size, which means that these markers are able to recover information from degraded DNA samples better than STRs that have amplicons as large as 300 bp to 400 bp. Second, they in principle can be multiplexed to a higher level than STRs because some detection methods (e.g., array hybridization) are not constrainted by electrophoretic space (e.g., 4 fluorescent dye labels and a size range of 75 bp to 450 bp). Third, the sample processing and data analysis can be more fully automated because a size-based separation is not needed. Fourth, there is no stutter artifact associated with each allele, which should help simplify interpretation of the allele call. Finally, the ability to predict ethnic origin and certain physical traits may be possible with careful selection of SNP markers. SNPs are most likely to impact forensic applications with this last reason—the ability to estimate ethnicity and to predict phenotypes (Butler et al. 2007).

The vast majority of SNPs are bi-allelic meaning that they have two possible alleles and therefore three possible genotypes. For example, if the alleles for a SNP locus are A and B

(where "A" could represent a C, T, A, or G nucleotide and "B" could also be a C, T, A, or G), then the three possible genotypes would be AA, BB, or AB. Mixture interpretation can present a challenge with SNPs because it may be difficult to tell the difference between a true heterozygote and a mixture containing two homozygotes or a heterozygote and a homozygote (Butler et al. 2007). The ability to obtain quantitative information from SNP allele calls is important when attempting to decipher mixtures.

At this time, one of the biggest challenges to using SNPs in forensic DNA typing applications is the inability to simultaneously amplify enough SNPs in robust multiplexes from small amounts of DNA. Because a single bi-allelic SNP by itself yields less information than a multi-allelic STR marker, it is necessary to analyze a larger number of SNPs in order to obtain a reasonable power of discrimination to define a unique profile. Progress is being made in the area of multiplex PCR amplification and assays capable of amplifying and analyzing more than 50 SNPs simultaneously have been demonstrated (Sanchez et al. 2006).

Since each SNP locus typically possesses only two possible alleles, more markers are needed to obtain a high discriminatory power than for STR loci that possess multiple alleles. Computational analyses have shown that on average 25 to 45 SNP loci are needed to yield equivalent random match probabilities as the 13 core STR loci (Chakraborty et al. 1999). Another study predicted that 50 SNPs possessing frequencies in the range of 20% to 50% for the minor allele can theoretically result in likelihood ratios similar to approximately 12 STR loci (Gill 2001).

The number of SNPs needed may fluctuate in practice because some SNP loci have variable allele frequencies in different population groups. Most likely a battery of 50 to 100 SNPs will be required to match the same powers of discrimination and mixture resolution capabilities now achieved with 10 to 16 STR loci (Gill et al. 2004).

## SNP Marker Categories and Applications

At the 2007 Congress of the International Society for Forensic Genetics held in Copenhagen, Denmark, an international panel of experts discussed SNP markers, technologies, and applications (Butler et al. 2008). The panel was moderated by Niels Morling (University of Copenhagen, Denmark) and John Butler (NIST, USA) and consisted of Bruce Budowle (FBI, USA), Peter Gill (Forensic Science Service, UK), Ken Kidd (Yale University, USA), Chris Phillips (University of Santiago de Compostela, Spain), Peter Schneider (University of Cologne, Germany), and Peter Vallone (NIST, USA). At this meeting, SNP markers were categorized based on application.

SNP markers may be classified into four general uses (Budowle & van Daal 2008, Butler et al. 2008): (1) individual identification or identity testing SNPs, (2) lineage-informative SNPs, (3) ancestry-informative SNPs, and (4) phenotype-informative SNPs. Table 12.2 defines these classifications and lists references to several examples for each. Most of the work to date has been performed with the identity testing SNPs using autosomal loci. Some SNP detection assays have also been used for pathology and toxicology purposes (Kiehne & Kauferstein 2007). Work with lineage informative SNPs will also be covered in Chapters 13, 14, and 15 on Y-chromosome, mitochondrial DNA, and X-chromosome, respectively.

**TABLE 12.2** Categories of SNP Markers (See Budowle & van Daal 2008, Butler et al. 2008).

| Category | Characteristics | Examples |
|---|---|---|
| Identity SNPs<br>Individual Identification SNPs (IISNPs) | SNPs that collectively give very low probabilities of two individuals having the same multi-locus genotype | FSS 21plex (Dixon et al. 2005)<br>SNPforID 52plex (Sanchez et al. 2006)<br>Kidd group SNPs (Pakstis et al. 2010) |
| Lineage SNPs<br>Lineage Informative SNPs (LISNPs) | Sets of tightly linked SNPs that function as multi-allelic markers that can serve to identify relatives with higher probabilities than simple bi-allelic SNPs | mtDNA coding region SNPs (Coble et al. 2004)<br>Japanese Y-SNPs (Mizuno et al. 2010)<br>Haplotype blocks (Ge et al. 2010) |
| Ancestry SNPs<br>Ancestry Informative SNPs (AISNPs) | SNPs that collectively give a high probability of an individual's ancestry being from one part of the world or being derived from two or more areas of the world | SNPforID 34plex (Phillips et al. 2007b)<br>24 SNPs (Lao et al. 2010)<br>FSS YSNPs (Wetton et al. 2005) |
| Phenotype SNPs<br>Phenotype Informative SNPs (PISNPs) | SNPs that provide a high probability that the individual has particular phenotypes, such as a particular skin color, hair color, eye color, etc. | Red hair (Grimes et al. 2001)<br>"Golden" gene pigmentation (Lamason et al. 2005)<br>IrisPlex eye color (Walsh et al. 2010) |

## SNP Markers and Multiplexes

SNPs occur in noncoding regions of the genome as well as in genes (both exons and introns). Depending on the potential application, SNP markers might be selected outside of exons or the direct cause of a genetic mutation. From a forensic historical perspective, the first work with SNP typing was with the HLA-DQA1 and AmpliType PM kits (Butler 2010, *Fundamentals*, Chapter 3). These early PCR assays were sensitive but not very informative because collectively DQA1 and PolyMarker only examined six loci. In recent years, a great deal of work has been performed to characterize a number of new SNP loci and to develop useful SNP multiplex assays.

Several members of the European forensic DNA typing community launched a project in 2003 known as SNP*for*ID that worked to develop SNP assays to directly aid forensic DNA analysis. This group selected several sets of potential forensic SNP markers and developed highly multiplexed SNP assays. Population data was also gathered to measure SNP allele frequencies in various groups of interest. The SNP*for*ID website (http://www.snpforid.org) contains links to their publications and population data. The SNP*for*ID consortium has published a 52plex PCR and SNaPshot assay (Sanchez et al. 2006) as well as collaborated with Applied Biosystems to develop a 48 autosomal SNP + amelogenin sex-typing assay based on the GenPlex genotyping system (Tomas et al. 2008). The SNP*for*ID group has also developed a 34plex with ancestry informative SNPs (Phillips et al. 2007b).

Ken Kidd's group at Yale University has characterized allele frequencies for more than 100 SNPs in roughly 40 different global human populations in an effort to find optimal human identity SNP markers. This group has also published criteria for selecting what they term a "universal individual identification panel" of SNPs (Kidd et al. 2006, Pakstis et al. 2007, Pakstis et al. 2010).

### SNP Information Databases

Large national and international efforts have been underway over the past few years to catalog human variation found in the form of SNP markers. The SNP Consortium (TSC) was established in the spring of 1999 to create a high-density SNP map of the human genome. The TSC effort (http://snp.cshl.org/) produced several million mapped and characterized human SNP markers that have been entered into public databases including dbSNP housed at the National Institutes of Health's National Center for Biotechnology Information (http://www.ncbi.nlm.nih.gov/SNP).

The original TSC work became the foundation for the International HapMap project that was a follow-on to the Human Genome Project. The HapMap work involved typing 270 individuals from African, European, and Asian populations with several million SNPs (http://www.hapmap.org/). With these large ventures ongoing around the world, there is no shortage of available SNP markers and accompanying population data.

The University of Santiago de Compostela in Spain has developed a web interface called SPSmart (SNPs for Population Studies-smart engine) to evaluate SNP population data (Amigo et al. 2008). This website also has a browser for examining SNP allele frequencies from the SNP*for*ID 52plex and 34plex assays (http://spsmart.cesga.es/snpforid.php).

## SNP ANALYSIS TECHNIQUES

A number of SNP typing methods are available, each with its own strengths and weaknesses. Several reviews of SNP typing technologies have been published and can be consulted for a more in-depth view of methodologies (Gut 2001, Kwok 2001, Syvanen 2001, Sobrino et al. 2005, Perkel 2008). A few SNP analysis techniques are summarized in Table 12.3.

One of the important characteristics of a SNP assay is its ability to examine multiple markers simultaneously since SNPs are not as variable as STRs and typically a limited amount of DNA template is available in forensic casework. While pyrosequencing and TaqMan assays are limited in their multiplexing capabilities, Luminex and minisequencing (SNaPshot) assays enable multiplexed analysis of a dozen or more SNP markers simultaneously. A number of SNP assays for detecting mtDNA, Y-chromosome, and autosomal markers have been described in the literature using the SNaPshot approach, which can be run on ABI 310 and 3100 instruments already in use in most forensic DNA laboratories.

### SNaPshot: A Primer Extension Assay Capable of Multiplex Analysis

Minisequencing, sometimes referred to as SNaPshot, involves allele-specific primer extension with fluorescent dye-labeled dideoxynucleotide triphosphates (ddNTPs) to help visualize the results. There are three primary steps in performing minisequencing: amplification, primer extension, and analysis (Figure 12.1). First, the region around each SNP locus is amplified using PCR. Amplicons can be pooled following singleplex PCR or simultaneously generated using multiplex PCR. The remaining dNTPs and primers following PCR are destroyed by simply adding two different enzymes to the initial reaction tube or well. Exonuclease (Exo) chews up the single-stranded primers while shrimp alkaline phosphatase

**TABLE 12.3** SNP Analysis Techniques (See Perkel 2008 for Additional Technologies).

| Method | Description | References |
|---|---|---|
| Reverse dot blot or linear arrays | A series of allele-specific probes are attached to a nylon test strip at separate sites; biotinylated PCR products hybridize to their complementary probes and are then detected with a colorimetric reaction and evaluated visually. | Saiki et al. (1989), Reynolds et al. (2000) |
| Genetic bit analysis | Primer extension with ddNTPs is detected with a colorimetric assay in a 96-well format. | Nikiforov et al. (1994) |
| Direct sequencing | PCR products are sequenced and compared to reveal SNP sites. | Kwok et al. (1994) |
| Denaturing HPLC | Two PCR products are mixed and injected on an ion-paired reversed-phase HPLC; single base differences in the two amplicons will be revealed by extra heteroduplex peaks. | Hecker et al. (1999) |
| TaqMan 5′ nuclease assay | A fluorescent probe consisting of reporter and quencher dyes is added to a PCR reaction; amplification of a probe-specific product causes cleavage of the probe and generates an increase in fluorescence. | Livak (1999) |
| Fluorescence polarization | Primer extension across the SNP site with dye-labeled ddNTPs; monitoring changes in fluorescence polarization reveals which dye is bound to the primer. | Chen et al. (1999) |
| Mass spectrometry | Primer extension across the SNP site with ddNTPs; mass difference between the primer and extension product is measured to reveal nucleotide(s) present. | Haff & Smirnov (1997), Li et al. (1999) |
| High-density arrays (Affymetrix chip) | Thousands of oligonucleotide probes are represented at specific locations on a microchip array; fluorescently labeled PCR products hybridize to complementary probes to reveal SNPs. | Wang et al. (1998), Sapolsky et al. (1999) |
| Electronic dot blot (Nanogen chip) | Potential SNP alleles are placed at discrete locations on a microchip array; an electric field at each point in the array is used to control hybridization stringency. | Sosnowski et al. (1997), Gilles et al. (1999) |
| Molecular beacons | Hairpin stem on oligonucleotide probe keeps fluorophore and its quencher in contact until hybridization to DNA target, which results in fluorescence. | Giesendorf et al. (1998) |
| Oligonucleotide ligation assay (OLA) | Colorimetric assay in microtiter 96-well format involving ligation of two probes if the complementary base is present. | Delahunty et al. (1996) |
| $T_m$-shift genotyping | Allelic-specific PCR is performed with a GC-tail attached to one of the forward allele-specific primers; amplified allele with GC-tailed primer will exhibit a melting curve at a higher temperature. | Germer & Higuchi (1999) |
| Pyrosequencing | Sequencing by synthesis of 20–30 nucleotides beyond primer site; dNTPs are added in a specific order and those incorporated result in release of pyrophosphate and light through an enzyme cascade. | Ahmadian et al. (2000), Andreasson et al. (2002) |
| Allele-specific hybridization (Luminex 100) | Dye-labeled PCR products hybridize to oligonucleotide probes (representing the various SNP types) attached to as many as 100 different colored beads; each bead is interrogated to determine its color and whether or not a PCR product is attached as the beads pass two lasers in a flow cytometer. | Armstrong et al. (2000), Budowle et al. (2004) |

*(Continued)*

**TABLE 12.3** SNP Analysis Techniques (See Perkel 2008 for Additional Technologies). *(Continued)*

| Method | Description | References |
|---|---|---|
| Minisequencing (SNaPshot assay) | Allele-specific primer extension across the SNP site with fluorescently labeled ddNTPs; mobility modifying tails can be added to the 5′-end of each primer in order to spatially separate them during electrophoresis. | Tully et al. (1996) |
| SNPstream UHT | High-tech version of genetic bit analysis with a 384-well tag array and 12plex PCR. | Bell et al. (2002) |

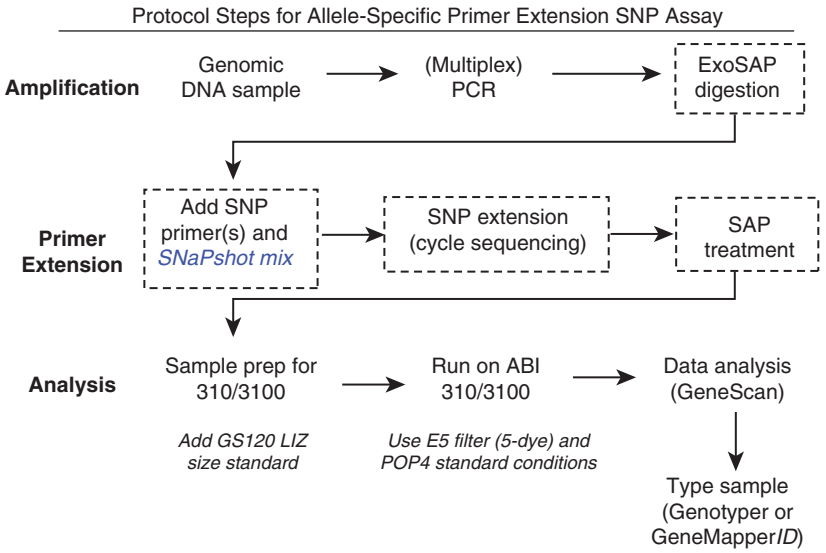
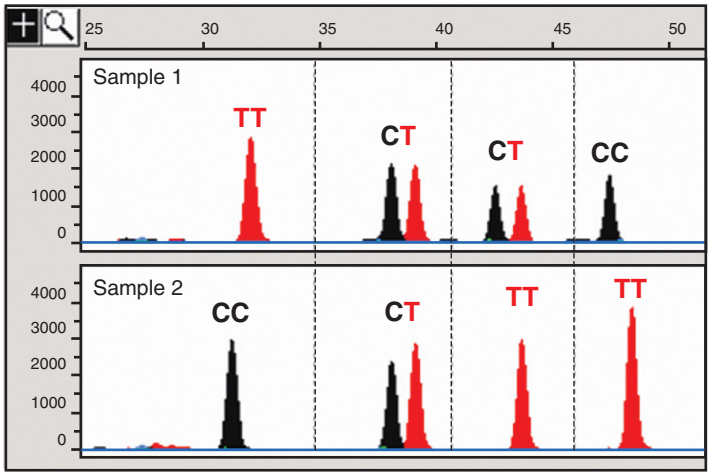



**FIGURE 12.1** Steps in allele-specific primer extension SNP detection (e.g., minisequencing or SNaPshot) assay. The boxed portions illustrate additional steps performed in this assay relative to STR typing.

(SAP) destroys the dNTP building blocks. These enzymes are often sold together as "ExoSAP." It is necessary to remove the primers and dNTPs so they do not interfere with the subsequent primer extension reaction.

Primer extension is performed by adding SNP extension primers, a mixture of the four possible ddNTPs each with a unique fluorescent dye label, and a polymerase to the ExoSAP-treated PCR products. The SNaPshot "kit" from Applied Biosystems only supplies the fluorescently labeled ddNTPs, buffer, and polymerase making it generic to any primer set. The SNP extension primers are designed to anneal immediately adjacent to a SNP site so that the addition of a single ddNTP will interrogate the nucleotide present at the SNP site in the PCR product. The SNP extension reaction is heated and cooled, usually through 25 cycles on a thermal cycler, to permit a linear amplification of the fluorescent ddNTP addition to the SNP primer by the polymerase. If any dNTPs remain from the preceding PCR reaction, then extension can go beyond the single base. Likewise, the presence of remaining

(a)




(b) (TTTTT)–**primer1** (chromosome 20)-**ddT/ddT**

(TTTTT)–(TTTTT)–**primer2** (chromosome 6)-**ddC/ddT**

(TTTTT)–(TTTTT)–(TTTTT)–**primer3** (chromosome 14)-**ddC/ddT**

(TTTTT)–(TTTTT)–(TTTTT)–(TTTTT)–**primer4** (chromosome 1)-**ddC/ddC**

FIGURE 12.2 Allele-specific primer extension results using four autosomal SNP markers on two different samples (a). SNP loci are from separate chromosomes (1, 6, 14, and 20) and therefore unlinked. Electrophoretic resolution of the SNP primer extension products occurs due to poly(T) tails that are 5 nucleotides different from one another (b).

PCR primers could mean that competing side reactions occur and thus interfere with the desired primer extension of the SNP primer.

Following the SNP extension reaction, the products are treated with SAP to remove unincorporated fluorescent ddNTPs. If the SAP treatment is incomplete, then dye artifacts (a.k.a. dye blobs) may occur in the electropherogram and obscure the SNP allele peaks being measured.

The availability of five-dye detection with electrophoretic platforms (see Chapter 6) enables an internal size standard to be added in the fifth dye channel to correct for run-to-run migration differences. Each of the four nucleotides has their own dye color: A (green), G (blue), C (yellow; usually displayed as black for better visual contrast), and T (red). Thus, the presence of a blue peak in the electropherogram would indicate that a G (ddGTP) had been incorporated by the polymerase at the SNP site.

Multiple primers can be analyzed simultaneously by linking a variable number of additional nucleotides to the 5′-end of the primers so that each primer differs by several nucleotides from its neighbor. Typically a poly(T) tail is used with a 3 to 5 base spread between primers (Tully et al. 1996, Vallone et al. 2004) although a mixed sequence that is not complementary to any human sequences has been used successfully (Sanchez et al. 2003). Thus, primer1 may contain a 5T 5′-tail, primer2 a 10T tail, primer3 a 15T tail, and primer4 a 20T tail in order to adequately resolve each locus during an electrophoretic separation (Figure 12.2).

The color of a peak in a minisequencing assay conveys the nucleotide present at the SNP site of interest while the size position of a peak correlates back to its locus based on the 5′-tailing used to differentiate the SNP marker from its neighbors. Homozygous alleles appear as single peaks and heterozygous alleles as two adjacent peaks (see upper middle section in Figure 12.2). Automated allele designation can be performed with computer programs such as Genotyper or GeneMapper designed to look for peaks in a particular color and size range.

A major advantage of allele-specific primer extension is that samples can be run on multicolor fluorescence detection electrophoresis instrumentation already available in most forensic DNA typing laboratories such as the ABI 310 or 3100 (see Chapter 6). The technique is sensitive, has the ability to be multiplexed, and has proven to be fairly robust with casework samples.

### High-Density SNP Arrays

The genomics community has graduated to next-generation sequencing technologies and high-throughput SNP typing to feed their appetite for more genetic information. Will the forensic community follow? Perhaps—if prices per sample drop and the vast amounts of data generated with these technologies become more manageable.

High-density SNP arrays permit hundreds of thousands or even millions of SNPs to be analyzed in parallel. However, these arrays typically require hundreds of nanograms of DNA, which is often not available from casework samples arising from minute biological stains. These SNP arrays are being used to perform genome-wide association studies (GWAS) to associate genetic variants with physical traits and disease markers. One of the side benefits of these studies in the identification of SNPs that could aid forensic phenotyping.

Direct-to-consumer companies, such as 23andMe (https://www.23andme.com/), are using high-density SNP arrays to provide genetic testing for biogeographical ancestry assessment and disease prediction using GWAS information collected from the growing scientific literature. These companies are claiming that kinship associations can be extended to second or even third degree relatives with a large number of SNPs—something that cannot be done with STR markers. In May 2009, Casework Genetics LLC (http://www.caseworkgen.com/) was founded with the goal of bringing the power of high-density SNP array technology to the forensic community. Using deconvolution algorithms (Homer et al. 2008), Casework Genetics hopes that massively parallel SNP results can help solve the mixture resolution issue previously mentioned for bi-allelic SNP markers.

## APPLICATIONS

In the following sections, each of the primary SNP applications will be briefly explored. References are provided at the back of the chapter to give interested readers further information on each topic.

### Identity SNPs—Supplementing Human Identity Testing

While SNPs will likely not replace STRs for DNA databasing and casework anytime soon (see Butler et al. 2007), researchers have identified a number of useful candidate loci for

human identity testing (Lee et al. 2005, Fang et al. 2009, Pakstis et al. 2010). Multiplex SNP assays have been constructed and population samples analyzed to measure allele frequencies (Vallone et al. 2005, Krjutskov et al. 2009, Nakahara et al. 2009, Pietrangeli et al. 2010).

As mentioned earlier, the SNP*for*ID project produced a 52plex PCR and SNP assay examining 52 autosomal SNP loci (Table 12.4). This assay, or minor variations of it, has been extensively studied in various population groups (Drobnic et al. 2010) and validated for forensic applications (Musgrave-Brown et al. 2007, Børsting et al. 2009) and paternity testing purposes (Børsting et al. 2008). These loci have also been tested for the NIST Standard Reference Material 2391b PCR-based DNA Profiling Standard (Børsting et al. 2011).

SNPs can serve as supplemental markers when STR results are not definitive enough in complex relationship tests (Phillips et al. 2008, Børsting & Morling 2011). The low mutation rate of SNPs relative to STRs means that the SNPs can help address multi-generational questions where STR mutations can produce an ambigous result. The Netherlands Forensic Institute has also developed tri-allelic SNPs to aid mixture interpretation and analysis of degraded DNA (Westen et al. 2009).

## Lineage SNPs—Aiding Kinship Analyses

Closely spaced autosomal SNPs are genetically linked (i.e., recombination does not occur between them to shuffle the alleles being passed on to the next generation). These linked SNPs can move as a haplotype block through generational transmission. In this form, these blocks of information provide more allelic states than do single loci and as such can be helpful for kinship analyses (Ge et al. 2010). Haplotype blocks also occur on the X-chromosome, the Y-chromosome, and mtDNA.

As will be discussed in the next three chapters, Y-chromosome, X-chromosome, and mitochondrial DNA markers can serve a variety of purposes in human identity testing. They are lineage markers because they are uniparentally transmitted. While STR markers are most commonly used for X and Y chromosome analysis, SNP loci can be beneficial in some circumstances due to their low mutation rate. For mtDNA, coding region SNPs can improve the ability to resolve common haplotypes (Coble et al. 2004). The forensic usefulness of a 25 X-SNP marker set has been explored (Tomas et al. 2010). Y-SNPs are routinely used to define haplogroups (Sanchez et al. 2003, Vallone & Butler 2004) and in some situations can aid biogeographical ancestry prediction (Brion et al. 2005, Wetton et al. 2005).

## Ancestry SNPs—Estimating Ethnicity

While many more SNPs than STRs are required to obtain similar random match probabilities, SNPs have the potential to be used in other ways to aid investigations, such as predicting a perpetrator's ancestral background. SNPs change on the order of once every $10^8$ generations while STR mutation rates are approximately one in a thousand. Because of this much lower mutation rate, SNPs are more likely to become "fixed" in a population than are STRs. SNPs are thus usually the better predictors of ethnicity (Barnholtz-Sloan et al. 2005). SNPs and *Alu* insertions (Batzer et al. 1993) are often found to be population-specific (Cordaux et al. 2007).

**TABLE 12.4** SNP*for*ID Loci Used in 52plex Assay (Sanchez et al. 2006). The Last Three Columns Show Allele Frequencies for the Various SNP Alleles in the Population Groups.

| No. | NCBI dbSNP ID | Chr. | Nucleotide Position | Amplicon Size (bp) | SNP Allele | African-American | Asian | European |
|---|---|---|---|---|---|---|---|---|
| 1 | rs1490413 | **1** | 4,037,521 | 68 | A/**G** | 0.59/0.41 | 0.46/0.54 | 0.47/0.53 |
| 2 | rs876724 | **2** | 104,974 | 83 | **C**/T | 0.72/0.28 | 0.46/0.54 | 0.69/0.31 |
| 3 | rs1357617 | **3** | 936,782 | 90 | **A**/T | 0.85/0.15 | 0.80/0.20 | 0.65/0.35 |
| 4 | rs2046361 | **4** | 10,719,942 | 79 | **A**/T | 0.65/0.35 | 0.70/0.30 | 0.80/0.20 |
| 5 | rs717302 | **5** | 2,932,133 | 86 | A/**G** | 0.92/0.08 | 0.83/0.17 | 0.45/0.55 |
| 6 | rs1029047 | **6** | 1,080,939 | 100 | A/T | 0.61/0.39 | 0.27/0.73 | 0.37/0.63 |
| 7 | rs917118 | **7** | 4,201,341 | 87 | C/**T** | 0.10/0.90 | 0.50/0.50 | 0.50/0.51 |
| 8 | rs763869 | **8** | 1,363,017 | 100 | C/**T** | 0.48/0.52 | – | 0.56/0.44 |
| 9 | rs1015250 | **9** | 1,813,774 | 95 | **C**/G | 0.55/0.45 | 0.40/0.60 | 0.15/0.85 |
| 10 | rs735155 | **10** | 3,328,178 | 100 | A/**G** | 0.38/0.62 | 0.82/0.18 | 0.41/0.59 |
| 11 | rs901398 | **11** | 11,060,530 | 70 | C/**T** | 0.39/0.61 | 0.29/0.71 | 0.34/0.66 |
| 12 | rs2107612 | **12** | 741,262 | 93 | **A**/G | 0.64/0.36 | 0.75/0.25 | 0.70/0.30 |
| 13 | rs1886510 | **13** | 20,172,700 | 86 | C/**T** | 0.85/0.15 | – | 0.50/0.50 |
| 14 | rs1454361 | **14** | 23,840,960 | 73 | **A**/T | 0.57/0.43 | 0.62/0.38 | 0.47/0.53 |
| 15 | rs2016276 | **15** | 22,119,157 | 90 | A/G | 0.91/0.09 | 0.59/0.41 | 0.79/0.21 |
| 16 | rs729172 | **16** | 5,606,490 | 60 | A/C | 0.23/0.77 | 0.06/0.94 | 0.38/0.62 |
| 17 | rs740910 | **17** | 5,907,188 | 87 | **A**/G | 0.92/0.08 | 0.92/0.09 | 0.57/0.43 |
| 18 | rs1493232 | **18** | 1,117,986 | 59 | A/**C** | 0.44/0.56 | 0.39/0.61 | 0.56/0.44 |
| 19 | rs719366 | **19** | 33,155,177 | 105 | C/**T** | 0.24/0.76 | 0.15/0.85 | 0.38/0.62 |
| 20 | rs1031825 | **20** | 4,442,483 | 98 | A/**C** | 0.60/0.40 | 0.41/0.59 | 0.30/0.70 |
| 21 | rs722098 | **21** | 15,607,469 | 81 | A/**G** | 0.17/0.83 | 0.50/0.50 | 0.86/0.14 |
| 22 | rs733164 | **22** | 26,141,338 | 68 | A/**G** | 0.26/0.74 | 0.16/0.84 | 0.34/0.66 |
| 23 | rs826472 | **10** | 2,360,631 | 85 | **C**/T | 0.70/0.30 | 0.66/0.34 | 0.44/0.56 |
| 24 | rs2831700 | **21** | 28,601,558 | 62 | A/G | 0.42/0.58 | 0.50/0.50 | 0.62/0.38 |
| 25 | rs873196 | **14** | 96,835,572 | 63 | **C**/T | 0.30/0.70 | 0.19/0.81 | 0.37/0.63 |
| 26 | rs1382387 | **16** | 79,885,888 | 69 | **G**/T | 0.55/0.45 | 0.40/0.60 | 0.25/0.75 |
| 27 | rs2111980 | **12** | 104,830,721 | 72 | **A**/G | 0.69/0.31 | 0.60/0.40 | 0.56/0.44 |
| 28 | rs2056277 | **8** | 139,370,172 | 73 | C/**T** | 0.70/0.30 | 0.75/0.25 | 0.55/0.45 |
| 29 | rs1024116 | **18** | 73,559,363 | 76 | A/**G** | 0.37/0.63 | 0.12/0.88 | 0.50/0.50 |
| 30 | rs727811 | **6** | 164,954,622 | 78 | A/**C** | 0.30/0.70 | 0.66/0.34 | 0.61/0.39 |

*(Continued)*

**TABLE 12.4** SNP*for*ID Loci Used in 52plex Assay (Sanchez et al. 2006). The Last Three Columns Show Allele Frequencies for the Various SNP Alleles in the Population Groups. *(Continued)*

| No. | NCBI dbSNP ID | Chr. | Nucleotide Position | Amplicon Size (bp) | SNP Allele | African-American | Asian | European |
|---|---|---|---|---|---|---|---|---|
| 31 | rs1413212 | **1** | 239,753,521 | 84 | **A**/G | 0.61/0.39 | 0.47/0.53 | 0.41/0.59 |
| 32 | rs938283 | **17** | 78,065,617 | 85 | **C**/T | 0.09/0.91 | 0.17/0.83 | 0.30/0.70 |
| 33 | rs1979255 | **4** | 191,013,970 | 86 | C/**G** | 0.38/0.62 | 0.33/0.67 | 0.35/0.65 |
| 34 | rs1463729 | **9** | 122,257,493 | 87 | **A**/G | 0.79/0.21 | 0.48/0.52 | 0.56/0.44 |
| 35 | rs2076848 | **11** | 134,205,198 | 89 | A/**T** | 0.38/0.62 | 0.58/0.42 | 0.49/0.51 |
| 36 | rs1355366 | **3** | 192,127,021 | 90 | A/**G** | 0.40/0.60 | 0.95/0.05 | 0.70/0.30 |
| 37 | rs907100 | **2** | 239,850,329 | 91 | C/G | 0.74/0.26 | – | 0.48/0.52 |
| 38 | rs354439 | **13** | 104,636,412 | 93 | **A**/T | 0.50/0.50 | 0.58/0.42 | 0.64/0.36 |
| 39 | rs2040411 | **22** | 46,048,653 | 94 | **A**/G | 0.75/0.25 | 0.08/0.92 | 0.71/0.29 |
| 40 | rs737681 | **7** | 154,850,085 | 96 | C/**T** | 0.56/0.44 | 0.80/0.20 | 0.58/0.42 |
| 41 | rs2830795 | **21** | 27,530,034 | 97 | **A**/G | 0.88/0.12 | 0.45/0.55 | 0.67/0.33 |
| 42 | rs251934 | **5** | 174,759,601 | 98 | **C**/T | 0.26/0.74 | 0.16/0.84 | 0.42/0.58 |
| 43 | rs914165 | **21** | 41,336,325 | 100 | A/**G** | 0.66/0.34 | 0.27/0.73 | 0.49/0.51 |
| 44 | rs10495407 | **1** | 235,480,457 | 102 | A/**G** | 0.11/0.89 | 0.42/0.58 | 0.29/0.71 |
| 45 | rs1360288 | **9** | 124,344,108 | 103 | **C**/T | 0.70/0.30 | 0.50/0.50 | 0.50/0.50 |
| 46 | rs964681 | **10** | 132,172,819 | 106 | **C**/T | 0.37/0.63 | 0.23/0.77 | 0.41/0.59 |
| 47 | rs1005533 | **20** | 40,172,539 | 107 | **A**/G | 0.55/0.45 | 0.60/0.40 | 0.45/0.55 |
| 48 | rs8037429 | **15** | 51,332,965 | 108 | **C**/T | 0.58/0.42 | 0.42/0.58 | 0.62/0.38 |
| 49 | rs891700 | **1** | 236,923,075 | 109 | A/**G** | 0.53/0.47 | 0.53/0.47 | 0.57/0.43 |
| 50 | rs1335873 | **13** | 18,699,724 | 110 | **A**/T | 0.91/0.09 | 0.28/0.72 | 0.36/0.64 |
| 51 | rs1028528 | **22** | 46,574,531 | 113 | A/**G** | 0.35/0.65 | 0.65/0.35 | 0.74/0.26 |
| 52 | rs1528460 | **15** | 52,926,761 | 115 | C/T | 0.65/0.35 | 0.36/0.64 | 0.29/0.71 |

Individuals with mixed ancestral backgrounds may not possess the expected phenotypic characteristics (e.g., dark colored skin for African-Americans). Thus, results from genetic tests attempting to predict ethnic origin or ancestry should always be interpreted with caution and only in the context of other reliable evidence. For several years before it went out of business in 2008, a company named DNAPrint (Sarasota, Florida) provided a DNA test for estimating an individual's ethnic/racial background with a panel of SNPs (Frudakis 2008). DNAPrint targeted pigmentation and xenobiotic metabolism genes in their search for ancestrally informative SNPs (Frudakis et al. 2003a). Much of their work was based on the research efforts of Dr. Mark Shriver from Penn State University (State College, PA) who

looks for ancestry informative markers (AIMs) that possess alleles with large frequency differences between populations (Shriver & Kittles 2004).

Population-specific alleles have been found in both STR and SNP markers. The DNAPrint SNP typing approach was used to aid the investigation of an important Louisianna serial rapist case in 2003 demonstrating the forensic value of this type of approach (Frudakis 2008, pp. 599-603). While presently used, AIMs are not 100% accurate for predicting ancestral background of samples (and perhaps never will be).

Several studies have shown that by analyzing many, many SNP markers it is possible to correlate genetic results with geographic location within Europe (Lao et al. 2008, Novembre et al. 2008). Mapping the country of origin may work for some places in Europe where language differences can discourage significant population movement. However, in countries like the United States where movement of the population is more fluid, greater levels of admixture are expected—and thus genetic testing results would not be as likely to correlate strongly with geographic location. Moreover, Myles and coworkers (2009) found that portability of ancestry informative markers was limited. In other words, identification of the optimal SNPs to differentiate populations in one group of samples does not necessarily mean that those same SNPs will work well in another set of samples.

While research results in this area are interesting, they are probably not ready for prime-time use in forensic investigations. It is important to keep in mind that investigators and juries may have a hard time understanding probabilities from ancestry or phenotyping predictions using DNA results. Telling a detective that the individual who left a biological sample at a crime scene has an 80% chance of having blue eyes is unfamiliar territory when he or she typically associates a DNA result as being irrefutable evidence (in their minds essentially 100% probability of being from a specific individual). If ancestry prediction and forensic phenotyping are pursued, then expectations of individuals using the information will need to be managed.

## Phenotype SNPs—Predicting External Visible Traits

When the DNA profile obtained from an evidentiary sample does not match a known individual—from either a tested suspect or a search of offender or arrestee indices in a DNA database—then the case currently cannot be solved through a DNA result. Under these circumstances, investigators may try to collect more potential references as part of mass screens or expand the reach of the database with familial searching (see Appendix 2). If these efforts fail, a John Doe warrant may be issued based on the evidentiary DNA profile to stop the statute of limitations, if one exists, from expiring. However, this requires waiting for the perpetrator to be caught committing another crime and then having his DNA profile submitted to the DNA database for the connection to the first crime to be made.

Another potential approach when no matching individual can be found to an evidentiary DNA profile is to try to gain more information from the evidentiary sample itself. Efforts have been made over the years to predict appearance of an individual using information from the evidentiary DNA sample (Sundquist 2010). For example, the so-called "golden" gene within exon 3 of melanin index-associated gene SLC24A5 on chromosome 15 (dbSNP accession number rs1426654) impacts pigmentation (Lamason et al. 2005). Results from this single SNP are also very useful in ancestry estimation studies as fair skin typically correlates well with European ancestry.

In the earliest example of forensic phenotyping to predict a physical trait from examining a sample's DNA, the Forensic Science Service developed a simple test of the melanocortin 1 receptor (MC1R) gene that correlated well with red hair color (Grimes et al. 2001). More recently, an IrisPlex has been developed to help predict blue versus brown eye color (Walsh et al. 2010a). The IrisPlex assay uses 6 SNPs and has been extensively studied in terms of sensitivity and specificity as part of forensic validation (Walsh et al. 2010b).

As more and more information is uncovered about the nature and content of the human genome, genetic variants that code for additional phenotypic characteristics will be discovered. Perhaps SNP sites can be identified in the future that will correlate to facial features thus aiding investigators with information about the possible appearance of a perpetrator. However, due to the complexity of multigenic traits and outside factors such as aging and environment, it is unlikely that a few carefully chosen SNPs will present a foolproof picture of a sample's source. Prominent researchers in this field conclude that "predicting a complete facial image from a biological sample as would be desirable for police investigations will remain forensic science fiction for at least the near future" (Kayser & Schneider 2009). Research is continuing in the area of forensic phenotyping; it is hoped that it will provide beneficial information to investigations in the future.

## ETHICAL CONSIDERATIONS

The expanded capabilities with ancestry and phenotypic SNPs to learn more about a biological sample than just simple identifying information available with STR typing raises potential ethical concerns. Should the capabilities and vast space of DNA information beyond the present STR profile be permitted for use in forensic casework?

In 2010, the University of Pennsylvania launched a forensic DNA ethics website (http://www.forensicdnaethics.org/) to explore the interplay between the advances being made in genetic information and appropriate privacy concerns. News stories on the topic of forensic phenotyping are available as is information on laws and scientific research efforts in this area.

Manfred Kayser's research group at the Erasmus University Medical Center in Rotterdam, The Netherlands, is probably the most active in an effort to bring phenotypic prediction to forensic science. Professor Kayser and Peter Schneider from the Unversity of Cologne (Cologne, Germany) published a landmark 2009 review covering the motivations, scientific challenges, and ethical considerations of DNA-based prediction for externally visible characteristics, such as hair color, eye color, facial features, etc. (Kayser & Schneider 2009). As discussed in Chapter 1, the Erasmus University phenotyping effort has also led to an approach to estimating human age based on T-cell DNA rearrangements, which provided an age estimate to within $\pm 9$ years (Zubakov et al. 2010).

Professors Kayser and Schneider argue that traits such as hair color and eye color cannot be considered private information as they are known to everyone who has ever seen the individual. They further note that since forensic phenotyping—or what they term "external visible characteristics (EVCs) prediction"—is only performed on crime scene samples of unknown persons that no violation of privacy rights has to be considered (Kayser & Schneider 2009).

In a law review, Koops and Schellenkens argue that the sensitivity of the information revealed should dictate the level of phenotyping permitted in a forensic context. Terri Sundquist, in her *Promega Connections* article on the topic, notes that "The Netherlands is the only country that explicitly allows forensic phenotyping, within defined guidelines" (Sundquist 2010). Only EVCs can be used and only when the trait, such as hair color or eye color, contributes to the criminal investigation. Looking for genetic susceptibility to disease or a tendency for aggressive behavior is not permitted under Dutch law. However, in other parts of the world including the UK, forensic phenotyping is implicity allowed. Scientific research in this area continues to go forward—and so public discussion and perhaps legislative limits should be considered.

## INSERTION-DELETION POLYMORPHISMS

Another form of a bi-allelic (or di-allelic) polymorphism is an insertion-deletion or indel. An indel can be the insertion or deletion of a segment of DNA ranging from one nucleotide to hundreds of nucleotides, which is seen with an *Alu* insertion (see Butler 2010, *Fundamentals*, D.N.A. Box 15.2). The two alleles for di-allelic indels can simply be classified as "short" and "long." From a certain perspective, STR markers can be thought of as multi-allelic indels since the different alleles are typically insertions or deletions of a tandem repeat unit.

Most di-allelic indels exhibit allele-length differences of only a few nucleotides. James Weber and colleagues at the Marshfield Medical Research Foundation have characterized over 2000 bi-allelic indels in the human genome (Weber et al. 2002). A total of 71% of these indels possessed 2-, 3-, or 4-nucleotide length differences with only 4% having greater than a 16-nucleotide length difference. Allele frequencies for the short and long alleles have been measured in African, European, Japanese, and Native American populations.

Rui Pereira and colleagues in Portugal and Spain have developed a multiplex assay with 38 bi-allelic autosomal indel markers that can be detected on standard capillary electrophoresis platforms (Pereira et al. 2009a, 2009b). The PCR amplicons were all designed to be less than 160 bp to accommodate degraded DNA. With this assay, complete profiles could be obtained down to approximately 300 pg of DNA template. All of the indel markers were found to be polymorphic in 306 individuals from Angola, Mozambique, Portugal, Macau, and Taiwan and produced random match probabilities on the order of $10^{-14}$, which is similar to about 13 STRs (Pereira et al. 2009a). In some countries, QIAGEN (Hilden, Germany) offers the Investigator DIPlex kit that examines 30 insertion/deletion polymorphisms using the ABI Genetic Analyzer platforms.

## SUMMARY AND FINAL THOUGHTS

It is unlikely that SNPs (or even other STR markers) will completely replace core STRs in the near- or even medium-term future as the primary source of information used in criminal investigations. Rather than replacing the millions of profiles that exist in large national DNA databases (see Chapter 8) through re-typing convicted offender and casework samples with

new SNP markers, it is more likely that new STRs or SNPs will be slowly added to national DNA databases as technologies are proven and costs come down.

The primary selling point of SNPs in the forensic arena has been the ability to make small PCR products that can overcome challenges of strong PCR inhibitors or samples containing highly degraded material. Smaller PCR products should result in greater recovery of information from badly damaged samples. PCR products for SNP markers can be small because the target region is only a single nucleotide rather than an expandable array of 20 to 60 nucleotides as is present in tetra-nucleotide STRs with 5 to 15 repeats.

While the promise of SNPs in this area has been promoted for many years, studies comparing miniSTRs (see Chapter 10) with SNPs have shown no clear advantage of SNPs with most degraded DNA samples (Dixon et al. 2006, Fondevilla et al. 2008). These reduced-size STR assays have an advantage over SNPs in that more alleles exist to produce a higher power of discrimination. More importantly miniSTR assays offer compatibility with current DNA databases housing millions of DNA profiles comprised of core STR loci (see Chapter 8).

Thanks in large measure to technology being fueled by worldwide genomic sequencing efforts, SNP analysis has become less expensive and easier to perform on a variety of detection platforms (Sobrino et al. 2005) . Many technologies, such as capillary array electrophoresis and mass spectrometry are capable of performing analysis on both STR and SNP markers. Although at the end of 2010—when this chapter was written—SNP typing is not widely performed in forensic laboratories. SNP typing has been used to supplement partial STR results on highly degraded specimens (Fondevilla et al. 2008).

For SNPs to become more widely used in future forensic DNA testing, it is important that the field settle on common loci. STR typing was propelled forward when the Second Generation Multiplex was introduced and used to launch the United Kingdom's National DNA Database (see Chapter 8). Likewise, the selection of 13 core loci for the Combined DNA Index System (CODIS) aided standardization of information in the United States and around the world (see Chapter 5).

To aid in cataloging SNP loci of forensic interest and collating features of the markers in a common format, the U.S. National Institute of Standards and Technology has set up a forensic SNP website: http://www.cstl.nist.gov/biotech/strbase/SNP.htm. This website is intended to provide a resource to the community as further markers, assays, and technologies are developed for SNP analysis. With this information many loci can be compared and examined for their forensic value to aid in the selection of a consistent set of SNP loci for the community to use as their standard.

In the spring of 2004, the DNA working group of the European Network of Forensic Science Institutes (ENFSI) and the U.S.-based Scientific Working Group on DNA Analysis Methods (SWGDAM) jointly issued an assessment of whether SNPs will replace STRs in national DNA databases (Gill et al. 2004). Their conclusion was that "it is unlikely that SNPs will replace STRs as the preferred method of testing of forensic samples and database samples in the near to medium future." The assessment goes on to praise the capabilities of SNPs for specific purposes including mass disaster and paternity analysis and comments on the need for standardization of SNP loci used for human identity testing applications. More than seven years later, the message of this paper holds true—SNPs may have a role for specific applications but will not replace STRs as the primary workhorse of forensic DNA typing any time soon.

# READING LIST AND INTERNET RESOURCES

Butler, J. M. (2010). *Fundamentals of forensic DNA typing*. San Diego: Elsevier Academic Press.

## General Information

Brookes, A. J. (1999). The essence of SNPs. *Gene*, *234*, 177–186.

Ku, C. S., et al. (2010). The discovery of human genetic variations and their use as disease markers: Past, present and future. *Human Genetics*, *55*, 403–415.

*STRBase forensic SNP information (assays, markers, etc.).* <http://www.cstl.nist.gov/biotech/strbase/SNP.htm>.

Syvanen, A. C. (2001). Accessing genetic variation: Genotyping single nucleotide polymorphisms. *Nature Reviews Genetics*, *2*, 930–942.

## Pros and Cons of SNPs

Amorim, A., & Pereira, L. (2005). Pros and cons in the use of SNPs in forensic kinship investigation: A comparative analysis with STRs. *Forensic Science International*, *150*, 17–21.

Balding, D. J. (2006). A tutorial on statistical methods for population association studies. *Nature Reviews Genetics*, *7*, 781–791.

Butler, J. M., et al. (2007). STRs vs SNPs: Thoughts on the future of forensic DNA testing. *Forensic Science, Medicine, and Pathology*, *3*, 200–205.

Chakraborty, R., et al. (1999). The utility of short tandem repeat loci beyond human identification: Implications for development of new DNA typing systems. *Electrophoresis*, *20*, 1682–1996.

Gill, P. (2001). An assessment of the utility of single nucleotide polymorphisms (SNPs) for forensic purposes. *International Journal of Legal Medicine*, *114*, 204–210.

Gill, P., et al. (2004). An assessment of whether SNPs will replace STRs in national DNA databases–joint considerations of the DNA working group of the European Network of Forensic Science Institutes (ENFSI) and the Scientific Working Group on DNA Analysis Methods (SWGDAM). *Science and Justice*, *44*, 51–53.

Krawczak, M. (1999). Informativity assessment for biallelic single nucleotide polymorphisms. *Electrophoresis*, *20*, 1676–1681.

## SNP Marker Categories and Applications

Budowle, B., & van Daal, A. (2008). Forensically relevant SNP classes. *Biotechniques*, *44*, 603–610.

Butler, J. M., et al. (2008). Report on ISFG SNP panel discussion. *Forensic Science International: Genetics Supplement Series*, *1*, 471–472.

Kiehne, N., & Kauferstein, S. (2007). Mutations in the SCN5A gene: Evidence for a link between long QT syndrome and sudden death? *Forensic Science International: Genetics*, *1*, 170–174.

## Markers and Multiplexes

Batzer, M. et al. (1993). *Alu* repeats as markers for human population genetics. *Proceedings of the fourth international symposium on human identification* (pp. 49–57). Madison, Wisconsin: Promega Corporation.

Cordaux, R., et al. (2007). In search of polymorphic Alu insertions with restricted geographic distributions. *Genomics*, *90*, 154–158.

Giardina, E., et al. (2007). In silico and in vitro comparative analysis to select, validate and test SNPs for human identification. *BMC Genomics*, *8*, 457.

Kidd, K. K., et al. (2006). Developing a SNP panel for forensic identification of individuals. *Forensic Science International*, *164*, 20–32.

Nicklas, J. A., & Buel, E. (2008). A real-time multiplex SNP melting assay to discriminate individuals. *Journal of Forensic Sciences*, *53*, 1316–1324.

Pakstis, A. J., et al. (2007). Candidate SNPs for a universal individual identification panel. *Human Genetics*, *121*, 305–317.

Pakstis, A. J., et al. (2010). SNPs for a universal individual identification panel. *Human Genetics*, *127*, 315–324.

Phillips, C., et al. (2007). Evaluation of the Genplex SNP typing system and a 49plex forensic marker panel. *Forensic Science International: Genetics*, *1*, 180–185.

Sanchez, J. J., et al. (2006). A multiplex assay with 52 single nucleotide polymorphisms for human identification. *Electrophoresis*, *27*, 1713–1724.
Sanchez, J. J., et al. (2008). Forensic typing of autosomal SNPs with a 29 SNP-multiplex—results of a collaborative EDNAP exercise. *Forensic Science International: Genetics*, *2*, 176–183.
*SNPforID project*. <http://www.snpforid.org/>.
Tomas, C., et al. (2008). Typing of 48 autosomal SNPs and amelogenin with GenPlex SNP genotyping system in forensic genetics. *Forensic Science International: Genetics*, *3*, 1–6.

## *SNP Information Databases*

ALFRED. <http://alfred.med.yale.edu/>.
Amigo, J., et al. (2008). SPSmart: Adapting population based SNP genotype databases for fast and comprehensive web access. *BMC Bioinformatics*, *9*, 428.
Amigo, J., et al. (2008). The SNPforID browser: An online tool for query and display of frequency data from the SNP*for*ID project. *International Journal of Legal Medicine*, *122*, 435–440.
dbSNP. <http://www.ncbi.nlm.nih.gov/sites/entrez?db=snp>.
Park, J., et al. (2007). SNP@Ethnos: A database of ethnically variant single-nucleotide polymorphisms. *Nucleic Acids Research*, *35*, D711–D715.
Smigielski, E. M., et al. (2000). dbSNP: A database of single nucleotide polymorphisms. *Nucleic Acids Research*, *28*, 352–355.
*SNPforID browser*. <http://spsmart.cesga.es/snpforid.php>.
*The SNP Consortium (TSC) and now International HapMap Data*. <http://snp.cshl.org/>.
Thorisson, G. A., & Stein, L. D. (2003). The SNP Consortium website: Past, present and future. *Nucleic Acids Research*, *31*, 124–127.

## *SNP Analysis Techniques*

Ahmadian, A., et al. (2000). Single-nucleotide polymorphism analysis by pyrosequencing. *Analytical Biochemistry*, *280*, 103–110.
Andréasson, H., et al. (2002). Mitochondrial sequence analysis for forensic identification using pyrosequencing technology. *Biotechniques*, *32*, 124–133.
Armstrong, B., et al. (2000). Suspension arrays for high throughput, multiplexed single nucleotide polymorphism genotyping. *Cytometry*, *40*, 102–108.
Bell, P. A., et al. (2002). SNPstream UHT: Ultra-high throughput SNP genotyping for pharmacogenomics and drug discovery. *Biotechniques*, *June Suppl*, 70–77.
Budowle, B. (2004). SNP typing strategies. *Forensic Science International*, *146S*, S139–S142.
Budowle, B., et al. (2004). Single nucleotide polymorphisms and microarray technology in forensic genetics—development and application to mitochondrial DNA. *Forensic Science Review*, *16*, 21–36.
Chen, X., et al. (1999). Fluorescence polarization in homogeneous nucleic acid analysis. *Genome Research*, *9*, 492–498.
Chen, X., & Sullivan, P. F. (2003). Single nucleotide polymorphism genotyping: Biochemistry, protocol, cost and throughput. *Pharmacogenomics Journal*, *3*, 77–96.
Delahunty, C., et al. (1996). Testing the feasibility of DNA typing for human identification by PCR and an oligonucleotide ligation assay. *American Journal of Human Genetics*, *58*, 1239–1246.
Divne, A. M., & Allen, M. (2005). A DNA microarray system for forensic SNP analysis. *Forensic Science International*, *154*, 111–121.
Fan, J. -B., et al. (2003). Highly parallel SNP genotyping. *Cold Spring Harbor Symposia on Quantitative Biology*, *68*, 69–78.
Germer, S., & Higuchi, R. (1999). Single-tube genotyping without oligonucleotide probes. *Genome Research*, *9*, 72–78.
Giesendorf, B. A. J., et al. (1998). Molecular beacons: A new approach for semi-automated mutation analysis. *Clinical Chemistry*, *44*, 482–486.
Gilles, P. N., et al. (1999). Single nucleotide polymorphic discrimination by an electronic dot bot assay on semiconductor microchips. *Nature Biotechnology*, *17*, 365–370.
Gut, I. G. (2001). Automation in genotyping of single nucleotide polymorphisms. *Human Mutation*, *17*, 475–492.
Haff, L. A., & Smirnov, I. P. (1997). Single-nucleotide polymorphism identification assays using a thermostable DNA polymerase and delayed extraction MALDI-TOF mass spectrometry. *Genome Research*, *7*, 378–388.

Hecker, K. H., et al. (1999). Mutation detection by denaturing DNA chromatography using fluorescently labeled polymerase chain reaction products. *Analytical Biochemistry*, *272*, 156–164.
Heller, M. J., et al. (2000). Active microelectronic chip devices which utilize controlled electrophoretic fields for multiplex DNA hybridization and other genomic applications. *Electrophoresis*, *21*, 157–164.
Homer, N., et al. (2008). Resolving individuals contributing trace amounts of DNA to highly complex mixtures using high-density SNP genotyping microarrays. *PLoS Genetics*, *4*(8), e1000167.
Hiratsuka, M., et al. (2005). Forensic assessment of 16 single nucleotide polymorphisms analyzed by hybridization probe assay. *Tohoku Journal of Experimental Medicine*, *207*, 255–261.
Illumina. <http://www.illumina.com/>.
Kwok, P.-Y., et al. (1994). Comparative analysis of human DNA variations by fluorescence-based sequencing of PCR products. *Genomics*, *23*, 138–144.
Kwok, P.-Y. (2001). Methods for genotyping single nucleotide polymorphisms. *Annual Reviews of Genomics and Human Genetics*, *2*, 235–258.
Li, J., et al. (1999). Single nucleotide polymorphism determination using primer extension and time-of-flight mass spectrometry. *Electrophoresis*, *20*, 1258–1265.
Li, L., et al. (2006). SNP genotyping by multiplex amplification and microarrays assay for forensic application. *Forensic Science International*, *162*, 74–79.
Livak, K. J. (1999). Allelic discrimination using fluorogenic probes and the 5′ nuclease assay. *Genetic Analysis*, *14*, 143–149.
McKeown, B., et al. (2006). Intramolecular controls for the generation of clinical-quality SNP genotypes and the assessment of normal heterozygote imbalance. *Electrophoresis*, *27*, 1725–1731.
Nakahara, H., et al. (2010). Criterion values for multiplex SNP genotyping by the Invader assay. *Forensic Science International: Genetics*, *4*, 130–136.
Nikiforov, T. T., et al. (1994). Genetic Bit Analysis: A solid phase method for typing single nucleotide polymorphisms. *Nucleic Acids Research*, *22*, 4167–4175.
Perkel, J. (2008). SNP genotyping: Six technologies that keyed a revolution. *Nature Methods*, *5*, 447–453.
Reynolds, R., et al. (2000). Detection of sequence variation in the HVII region of the human mitochondrial genome in 689 individuals using immobilized sequence-specific oligonucleotide probes. *Journal of Forensic Sciences*, *45*, 1210–1231.
Saiki, R. K., et al. (1989). Genetic analysis of amplified DNA with immobilized sequence-specific oligonucleotide probes. *Proceedings of the National Academy of the United States of America*, *86*, 6230–6234.
Sapolsky, R. J., et al. (1999). High-throughput polymorphism screening and genotyping with high-density oligonucleotide arrays. *Genetic Analysis*, *14*, 187–192.
Sobrino, B., et al. (2005). SNPs in forensic genetics: A review on SNP typing methodologies. *Forensic Science International*, *154*, 181–194.
Sosnowski, R. G., et al. (1997). Rapid determination of single base mismatch mutations in DNA hybrids by direct electric field control. *Proceedings of the National Academy of the United States of America*, *94*, 1119–1123.
Tully, G., et al. (1996). Rapid detection of mitochondrial sequence polymorphisms using multiplex solid-phase fluorescent minisequencing. *Genomics*, *34*, 107–113.
Wang, D. G., et al. (1998). Large-scale identification, mapping, and genotyping of single-nucleotide polymorphisms in the human genome. *Science*, *280*, 1077–1082.
Ye, J., et al. (2002). Melting curve SNP (McSNP) genotyping: A useful approach for diallelic genotyping in forensic science. *Journal of Forensic Sciences*, *47*, 593–600.

### *SNP Applications*

Frudakis, T. N. (2008). *Molecular photofitting: Predicting ancestry and phenotype using DNA*. San Diego: Elsevier Academic Press.

### *Identity SNPs*

Børsting, C., et al. (2008). Performance of the SNPforID 52 SNP-plex assay in paternity testing. *Forensic Science International: Genetics*, *2*, 292–300.
Børsting, C., et al. (2009). Validation of a single nucleotide polymorphism (SNP) typing assay with 49 SNPs for forensic genetic testing in a laboratory accredited according to the ISO 17025 standard. *Forensic Science International: Genetics*, *4*, 34–42.

Børsting, C., & Morling, N. (2011). Mutations and/or close relatives? Six case work examples where 49 autosomal SNPs were used as supplementary markers. *Forensic Science International: Genetics*, *5*, 236–241.
Børsting, C., et al. (2011). SNP typing of the reference materials SRM 2391b 1-10, K562, XY1, XX74, and 007 with the SNPforID multiplex. *Forensic Science International: Genetics*, *5*, e81–e82.
Dixon, L. A., et al. (2005). Validation of a 21-locus autosomal SNP multiplex for forensic identification purposes. *Forensic Science International*, *154*, 62–77.
Drobnič, K., et al. (2010). Typing of 49 autosomal SNPs by SNaPshot in the Slovenian population. *Forensic Science International: Genetics*, *4*, e125–e127.
Fang, R., et al. (2009). Multiplexed SNP detection panels for human identification. *Forensic Science International: Genetics Supplement Series*, *2*, 538–539.
Fondevila, M., et al. (2008). Case report: Identification of skeletal remains using short-amplicon marker analysis of severely degraded DNA extracted from a decomposed and charred femur. *Forensic Science International: Genetics*, *2*, 212–218.
Hinds, D. A., et al. (2005). Whole-genome patterns of common DNA variation in three human populations. *Science*, *307*, 1072–1079.
Krjutskov, K., et al. (2009). Evaluation of the 124-plex SNP typing microarray for forensic testing. *Forensic Science International: Genetics*, *4*, 43–48.
Lee, H. Y., et al. (2005). Selection of twenty-four highly informative SNP markers for human identification and paternity analysis in Koreans. *Forensic Science International*, *148*, 107–112.
Musgrave-Brown, E., et al. (2007). Forensic validation of the SNP*for*ID 52-plex assay. *Forensic Science International: Genetics*, *1*, 186–190.
Nakahara, H., et al. (2009). Automated SNPs typing system based on the Invader assay. *Legal Medicine*, *11*(Suppl. 1), S111–S114.
Pakstis, A. J., et al. (2008). SNPs for individual identification. *Forensic Science International: Genetics Supplement Series*, *1*, 479–481.
Phillips, C., et al. (2008). Resolving relationship tests that show ambiguous STR results using autosomal SNPs as supplementary markers. *Forensic Science International: Genetics*, *2*, 198–204.
Pietrangeli, I., et al. (2010). Frequency assessment of 25 SNPs in five different populations. *Forensic Science International: Genetics*, *4*, e131–e133.
Sanchez, J. J., et al. (2006). A multiplex assay with 52 single nucleotide polymorphisms for human identification. *Electrophoresis*, *27*, 1713–1724.
Tomas, C., et al. (2010). Autosomal SNP typing of forensic samples with the GenPlex HID System: Results of a collaborative study. *Forensic Science International: Genetics* (in press) (doi:10.1016/j.fsigen.2010.06.007).
Vallone, P. M., et al. (2005). Allele frequencies for 70 autosomal SNP loci with U.S. Caucasian, African-American, and Hispanic samples. *Forensic Science International*, *149*, 279–286.
Westen, A. A., et al. (2009). Tri-allelic SNP markers enable analysis of mixed and degraded DNA samples. *Forensic Science International: Genetics*, *3*, 233–241.

## *Lineage SNPs*

Bouakaze, C., et al. (2007). First successful assay of Y-SNP typing by SNaPshot minisequencing on ancient DNA. *International Journal of Legal Medicine*, *121*, 493–499.
Brion, M., et al. (2005). Introduction of an single nucleotide polymorphism-based "major Y-chromosome haplogroup typing kit" suitable for predicting the geographical origin of male lineages. *Electrophoresis*, *26*, 4411–4420.
Coble, M. D., et al. (2004). Single nucleotide polymorphisms over the entire mtDNA genome that increase the power of forensic testing in Caucasians. *International Journal of Legal Medicine*, *118*, 137–146.
Ge, J., et al. (2010). Haplotype block: A new type of forensic DNA markers. *International Journal of Legal Medicine*, *124*, 353–361.
Kline, M. C., et al. (2005). Mitochondrial DNA typing screens with control region and coding region SNPs. *Journal of Forensic Sciences*, *50*, 377–385.
Mizuno, N., et al. (2010). A forensic method for the simultaneous analysis of biallelic markers identifying Y chromosome haplogroups inferred as having originated in Asia and the Japanese archipelago. *Forensic Science International: Genetics*, *4*, 73–79.
Nelson, T. M., et al. (2007). Development of a multiplex single base extension assay for mitochondrial DNA haplogroup typing. *Croatian Medical Journal*, *48*, 460–472.

Sanchez, J. J., et al. (2003). Multiplex PCR and minisequencing of SNPs—a model with 35 Y chromosome SNPs. *Forensic Science International*, *137*, 74–84.

Tomas, C., et al. (2010). Forensic usefulness of a 25 X-chromosome single-nucleotide polymorphism marker set. *Transfusion*, *50*, 2258–2265.

Vallone, P. M., et al. (2004). A multiplex allele-specific primer extension assay for forensically informative SNPs distributed throughout the mitochondrial genome. *International Journal of Legal Medicine*, *118*, 147–157.

Vallone, P. M., & Butler, J. M. (2004). Y-SNP typing of U.S. African American and Caucasian samples using allele-specific hybridization and primer extension. *Journal of Forensic Sciences*, *49*, 723–732.

### *Ancestry SNPs*

Barnholtz-Sloan, J. S., et al. (2005). Informativeness of the CODIS STR loci for admixture analysis. *Journal of Forensic Sciences*, *50*, 1322–1326.

Corach, D., et al. (2010). Inferring continental ancestry of Argentineans from autosomal, Y-chromosomal and mitochondrial DNA. *Annuals of Human Genetics*, *74*, 65–76.

Frudakis, T., et al. (2003). A classifier for the SNP-based inference of ancestry. *Journal of Forensic Sciences*, *48*, 771–782.

Halder, I., et al. (2008). A panel of ancestry informative markers for estimating individual biogeographical ancestry and admixture from four continents: Utility and applications. *Human Mutation*, *29*, 648–658.

Halder, I., et al. (2009). Measurement of admixture proportions and description of admixture structure in different U.S. populations. *Human Mutation*, *30*, 1299–1309.

Kersbergen, P., et al. (2009). Developing a set of ancestry-sensitive DNA markers reflecting continental origins of humans. *BMC Genetics*, *10*, 69.

Kosoy, R., et al. (2009). Ancestry informative marker sets for determining continental origin and admixture proportions in common populations in America. *Human Mutation*, *30*, 69–78.

Lao, O., et al. (2006). Proportioning whole-genome single-nucleotide–polymorphism diversity for the identification of geographic population structure and genetic ancestry. *American Journal of Human Genetics*, *78*, 680–690.

Lao, O., et al. (2008). Correlation between genetic and geographic structure in Europe. *Current Biology*, *18*, 1241–1248.

Lao, O., et al. (2010). Evaluating self-declared ancestry of U.S. Americans using autosomal, Y-chromosomal and mitochondrial DNA. *Human Mutation*, *31*, E1875–E1893.

Myles, S., et al. (2009). An assessment of the portability of ancestry informative markers between human populations. *BMC Medical Genomics*, *2*(45), 1–10.

Novembre, J., et al. (2008). Genes mirror geography within Europe. *Nature*, *456*, 98–101.

Phillips, C., et al. (2007). Inferring ancestral origin using a single multiplex assay of ancestry-informative marker SNPs. *Forensic Science International: Genetics*, *1*, 273–280.

Phillips, C., et al. (2009). Ancestry analysis in the 11-M Madrid bomb attack investigation. *PLoS ONE*, *4*(8), e6583.

Pfaff, C. L., et al. (2004). Information on ancestry from genetic markers. *Genetic Epidemiology*, *26*, 305–315.

Price, A. L., et al. (2008). Discerning the ancesty of European Americans in genetic association studies. *PLoS Genetics*, *4*(1), e236.

Schlecht, J., et al. (2008). Machine-learning approaches for classifying haplogroup from Y chromosome STR data. *PLoS Computational Biology*, *4*(6), e1000093.

Shriver, M. D., & Kittles, R. A. (2004). Genetic ancestry and the search for personalized genetic histories. *Nature Reviews Genetics*, *5*, 611–618.

Wetton, J. H., et al. (2005). Inferring the population of origin of DNA evidence within the UK by allele-specific hybridization of Y-SNPs. *Forensic Science International*, *152*, 45–53.

### *Phenotype SNPs*

Axenovich, T. I., et al. (2009). Linkage analysis of adult height in a large pedigree from a Dutch genetically isolated population. *Human Genetics*, *126*, 457–471.

Bastiaens, M., et al. (2001). The melanocortin-1 receptor gene is the major freckle gene. *Human Molecular Genetics*, *10*, 1701–1708.

Bouakaze, C., et al. (2009). Pigment phenotype and biogeographical ancestry from ancient skeletal remains: Inferences from multiplexed autosomal SNP analysis. *International Journal of Legal Medicine*, *123*, 315–325.

Branicki, W., et al. (2007). Determination of phenotype associated SNPs in the MC1R gene. *Journal of Forensic Sciences*, *52*, 349–354.
Duffy, D. L., et al. (2007). A three-single-nucleotide polymorphism haplotype in intron 1 of OCA2 explains most human eye-color variation. *American Journal of Human Genetics*, *80*, 241–252.
Frudakis, T., et al. (2003). Sequences associated with human iris pigmentation. *Genetics*, *165*, 2071–2083.
Graham, E. A. M. (2008). DNA reviews: Predicting phenotype. *Forensic Science, Medicine, and Pathology*, *4*, 196–199.
Grimes, E. A., et al. (2001). Sequence polymorphism in the human melanocortin 1 receptor gene as an indicator of the red hair phenotype. *Forensic Science International*, *122*, 124–129.
Kayser, M., et al. (2008). Three genome-wide association studies and a linkage analysis identify *HERC2* as a human iris color gene. *American Journal of Human Genetics*, *82*, 411–423.
Lalueza-Foc, C., et al. (2007). A melanocortin 1 receptor allele suggests varying pigmentation among Neanderthals. *Science*, *318*, 1453–1455.
Lamason, R. L., et al. (2005). SLC24A5, a putative cation exchanger, affects pigmentation in zebrafish and humans. *Science*, *310*, 1782–1786.
Lao, O., et al. (2007). Signatures of positive selection in genes associated with human skin pigmentation as revealed from analyses of single nucleotide polymorphisms. *Annuals of Human Genetics*, *71*, 354–369.
Liu, F., et al. (2009). Eye color and the prediction of complex phenotypes from genotypes. *Current Biology*, *19*, R192–R193.
Liu, F., et al. (2010). Digital quantification of human eye color highlights genetic association of three new loci. *PLoS Genetics*, *6*(5), e1000934. Available at <http://www.plosgenetics.org>.
Meissner, C., & Ritz-Timme, S. (2010). Molecular pathology and age estimation. *Forensic Science International: Genetics*, *203*, 34–43.
Mengel-From, J., et al. (2010). Human eye colour and *HERC2, OCA2,* and *MATP*. *Forensic Science International: Genetics*, *4*, 323–328.
Pulker, H., et al. (2007). Finding genes that underlie physical traits of forensic interest using genetic tools. *Forensic Science International: Genetics*, *1*, 100–104.
Schmeling, A., et al. (2007). Age estimation. *Forensic Science International*, *165*, 178–181.
Soares, C. (2010). Portrait in DNA: Can forensic analysis yield police-style sketches of suspects? How your genes could reveal what you look like. *Scientific American*(May 11, 2010 news). Available at <http://www.scientific-american.com/article.cfm?id=portrait-in-dna>.
Sturm, R. A., et al. (2008). A single SNP in an evolutionary conserved region within intron 86 of the *HERC2* gene determines human blue-brown eye color. *American Journal of Human Genetics*, *82*, 424–431.
Sturm, R. A. (2009). Molecular genetics of human pigmentation diversity. *Human Molecular Genetics*, *18*, R9–R17.
Sulem, P., et al. (2007). Genetic determinants of hair, eye and skin pigmentation in Europeans. *Nature Genetics*, *39*, 1443–1452.
Sundquist, T. (2010). Forensic phenotyping: What DNA can (and cannot) tell us about a criminal's appearance. *Promega Connections* (research blog April 26, 2010). Available at <http://promega.wordpress.com/>.
Tully, G. (2007). Genotype versus phenotype: Human pigmentation. *Forensic Science International: Genetics*, *1*, 105–110.
Valenzuela, R., et al. (2010). Predicting phenotype from genotype: Normal pigmentation. *Journal of Forensic Sciences*, *55*(2), 315–322.
Valverde, P., et al. (1995). Variants of the melanocyte-stimulating hormone receptor gene are associated with red hair and fair skin in humans. *Nature Genetics*, *11*, 328–330.
Walsh, S., et al. (2010). IrisPlex: A sensitive DNA tool for accurate prediction of blue and brown eye colour in the absence of ancestry information. *Forensic Science International: Genetics*, *5*, 170–180.
Walsh, S., et al. (2010). Developmental validation of the IrisPlex system: Determination of blue and brown iris colour for forensic intelligence. *Forensic Science International: Genetics* (in press). doi:10.1016/j/fsigen.2010.09.008
Zubakov, D., et al. (2010). Estimating human age from T-cell DNA rearrangements. *Current Biology*, *20*, R970–R971.

## *Ethical Considerations with Phenotype and Ethnicity Prediction*

*Forensic DNA ethics*. <http://forensicdnaethics.org/>.
Kayser, M., & Schneider, P. M. (2009). DNA-based prediction of human externally visible characteristics in forensics: Motivations, scientific challenges, and ethical considerations. *Forensic Science International: Genetics*, *3*, 154–161.

Koops, B.-J., & Schellekens, M. (2008). Forensic DNA phenotyping: Regulatory issues. *The Columbia Science and Technology Law Review*, *9*, 158–202. Available at <http://www.stlr.org/html/volume9/koops.pdf>.

### *Insertion/Deletion Polymorphisms*

Pereira, R., et al. (2009). A new multiplex for human identification using insertion/deletion polymorphisms. *Electrophoresis*, *30*, 3682–3690.

Pereira, R., et al. (2009). Insertion/deletion polymorphisms: A multiplex assay and forensic applications. *Forensic Science International: Genetics Supplement Series*, *2*, 513–515.

Santos, N. P. C., et al. (2010). Assessing individual interethnic admixture and population substructure using a 48-insertion-deletion (INDEL) ancestry-informative marker (AIM) panel. *Human Mutation*, *31*, 184–190.

Weber, J. L., et al. (2002). Human diallelic insertion/deletion polymorphisms. *American Journal of Human Genetics*, *71*, 854–862.